

Inyoung Ryu

Texas A&M University
Mathematics Department
riy520@tamu.edu

RESEARCH INTERESTS

- Surface group representations and Character variety
- Mapping class groups and Teichmüller space
- Low-dimensional topology and Geometric group theory

EDUCATION

Texas A&M University <i>Ph.D. in Mathematics</i>	Aug.2021 – Present <i>Texas, US</i>
Ulsan National Institute of Science and Technology(UNIST) <i>Bachelor of Science in Mathematical Science</i> • Minor in Computer Science and Engineering	Mar.2017 – Jul.2021 <i>Ulsan, Korea</i>
Sabancı University <i>Erasmus Program</i>	Feb. 2020 – Jun. 2020 <i>Istanbul, Turkey</i>

PUBLICATION/PREPRINTS

From Bowditch question to Goldman conjecture for type-preserving representations <i>with Viraj Joshi, submitted, arXiv: 2609.02065</i>	Sep. 2026
On totally hyperbolic non-Fuchsian type-preserving representations <i>Submitted, arXiv:2511.13989</i>	Nov. 2025
Connected components of spaces of type-preserving representations <i>with Tian Yang, submitted, arXiv: 2502.05078</i>	Jul. 2025

INVITED SEMINAR/COLLOQUIUM TALKS

(Upcoming) Two invariants of type-preserving representations and applications <i>Joint mathematics meetings 2027</i>	Jan. 2027 <i>Illinois, US</i>
(Upcoming) Two invariants of type-preserving representations and applications <i>Geometric Group Theory Seminar, Purdue University</i>	Nov. 2026 <i>Indiana, US</i>
Two invariants of type-preserving representations and applications <i>Geometry and Dynamics Seminar, Centro de Ciencias Matemáticas UNAM (Online)</i>	Sep. 2026 <i>Morelia, Mexico</i>
Two invariants of type-preserving representations and applications <i>Topology Seminar, University of Tennessee</i>	Apr. 2026 <i>Tennessee, US</i>
Two invariants of type-preserving representations and applications <i>Topology & Geometry Seminar, Rutgers University</i>	Apr. 2026 <i>New Jersey, US</i>
Two invariants of type-preserving representations and applications <i>Algebra & Combinatorics Seminar, Texas A&M University</i>	Mar. 2026 <i>Texas, US</i>
Two invariants of type-preserving representations and applications <i>Groups & Dynamics Seminar, University of Texas Austin</i>	Mar. 2026 <i>Texas, US</i>
Two invariants of type-preserving representations and applications <i>Groups & Dynamics Seminar, Texas A&M University</i>	Feb. 2026 <i>Texas, US</i>
Connected components of the space of type-preserving representations <i>SHSU Junior Colloquium, Sam Houston State University</i>	Nov. 2025 <i>Texas, US</i>
Connected components of the space of type-preserving representations <i>KIAS HCMC Topology Seminar, Korea Institute for Advanced Study (Online)</i>	Feb. 2025 <i>Seoul, Korea</i>
Geometric approach to surface group representations <i>Graduate Student Organization Seminar, Texas A&M University</i>	Sep. 2024 <i>Texas, US</i>

Topological components of spaces of representations*SNU Topology Seminar, Seoul National University (Online)*

Aug. 2024

Seoul, Korea

Mapping class group action on Teichmüller Space*Graduate Student Organization Seminar, Texas A&M University*

Apr. 2024

Texas, US

CONTRIBUTED TALKS**(Upcoming) On totally hyperbolic non-Fuchsian type-preserving representations***AMS 2026 Fall Western Sectional Meeting, Arizona State University*

Nov. 2026

Arizona, US

(Upcoming) Non-Fuchsian totally hyperbolic type-preserving representations*AMS 2026 Fall Central Sectional Meeting, University of Minnesota Duluth*

Oct. 2026

Minnesota, US

Connected components of the space of type-preserving representations*GTA: Philadelphia 2025, Temple University*

May. 2025

Pennsylvania, US

Connected components of the space of type-preserving representations*CIRM School on Geometric structures and discrete group actions*

Apr. 2025

Marseille, France

Connected components of the space of type-preserving representations*Graduate Student Conference in Topology and Geometry, Indiana University Bloomington*

Apr. 2025

Indiana, US

Nielsen-Thurston classification*GIGEM Conference, Texas A&M University*

Apr. 2024

Texas, US

LIGHTNING TALKS/POSTER PRESENTATION**Can a non-Fuchsian representation be totally hyperbolic?***Vanderbilt colloquium series: Groups, geometry, and dynamics (ICM satellite conference)*

Jul. 2026

Tennessee, US

Poster Presentation: Non-Fuchsian Totally hyperbolic Representations*Young Geometric Group Theory XIV*

Jun. 2026

Seoul, Korea

Connected components of the space of type-preserving representations*Ventotene Workshop on Higher dimensional hyperbolic geometry*

Sep. 2025

Ventotene, Italy

Connected components of the space of type-preserving representations*2K-GATE Workshop, Korea Institute for Advanced Study*

Jun. 2025

Seoul, Korea

Connected components of the space of type-preserving representations*53rd Barrett Lectures: Recent Development in Geometric Group Theory*

May. 2025

Tennessee, US

Connected components of the space of type-preserving representations*Conference on Geometry and Dynamics in Low Dimensions, IMVAST*

Jan. 2025

Hanoi, Vietnam

SELECTED CONFERENCE/WORKSHOPS**International Congress of Mathematicians (ICM 2026)**

Jul. 2026

Pennsylvania, US

Joint Mathematics Meeting (JMM 2026)

Jan. 2026

Washington, D.C., US

Workshop on Homogeneous Dynamics and Anosov Representations*Simons Laufer Mathematical Sciences Institute*

Apr. 2026

California, US

SLMath Summer School: Geometry and Dynamics in Higher Rank Lie Groups*University of California, Berkeley*

Jul. 2025

California, US

Workshop on Low-dimensional phenomena: geometry and dynamics*Institut henri poincaré*

Jun. 2025

Paris, France

Workshop on Lorentzian, Affine, and Hyperbolic Differential Geometry*University of Maryland*

Apr. 2025

Maryland, US

Workshop on Motives and Mapping Class Groups*American Institute of Mathematics, Caltech*

Jan. 2025

California, US

Joint Mathematics Meeting (JMM 2024)

Jan. 2024

California, US

Conference on Characters and Moduli of Surfaces*Kyoto University*

Jul. 2023

Kyoto, Japan

HONORS/AWARDS/GRANTS

AMS Fall 2026 Sectional Meeting Travel Grant <i>American Mathematical Society</i>	Sep. 2024 <i>Texas, US</i>
Houston A&M Mothers' Club Outstanding TA Award <i>Selected for going above and beyond expectations as a teaching assistant</i>	Jul. 2024 <i>Texas, US</i>

TEACHING

Graduate Research Assistant, supported by NSF Grant, P.I. - Tian Yang <i>Texas A&M University</i>	Sep. 2026 - Dec.2026 <i>Texas, US</i>
Python Lab Assignment, MATH 151/152 (Engineering Mathematics I/II) <i>Texas A&M University</i>	Jul.2026 <i>Texas, US</i>
Instructor of record, Math 142 (Business Calculus) <i>Texas A&M University</i>	Jan. 2026 - May. 2026 <i>Texas, US</i>
Grader, MATH 636 (Topology I) <i>Texas A&M University</i>	Sep. 2025 - Dec.2025 <i>Texas, US</i>
Python Lab Assignment, MATH 150 (Functions, Trigonometry and Linear Systems) <i>Texas A&M University</i>	Jul.2025 <i>Texas, US</i>
MATH 467 (Modern Geometry) Grader <i>Texas A&M University</i>	Jan. 2025 - May.2025 <i>Texas, US</i>
Graduate Research Assistant, supported by NSF Grant, P.I. - Tian Yang <i>Texas A&M University</i>	Sep. 2024 - Dec.2024 <i>Texas, US</i>
Graduate Research Assistant, supported by NSF Grant, P.I. - Tian Yang <i>Texas A&M University</i>	Jul. 2024 – Dec. 2024 <i>Texas, US</i>
Help Session Tutor, Math 142 (Business Calculus) <i>Texas A&M University</i>	Jun. 2024 <i>Texas, US</i>
Graduate Research Assistant, supported by NSF Grant, P.I. - Tian Yang <i>Texas A&M University</i>	Jan. 2024 – May. 2024 <i>Texas, US</i>
Grader, Math 636 (Topology I) <i>Texas A&M University</i>	Aug. 2023 – Dec. 2023 <i>Texas, US</i>
Grader, Math 142 (Business Calculus) <i>Texas A&M University</i>	Jul. 2023 <i>Texas, US</i>
Python Labs, Math 151(Engineering Math I) <i>Texas A&M University</i>	Jun. 2023 <i>Texas, US</i>
Teaching Assistant, Math 152(Engineering Math II) <i>Texas A&M University</i>	Jan. 2023 – May. 2023 <i>Texas, US</i>
Teaching Assistant, Math 151(Engineering Math I) <i>Texas A&M University</i>	Aug. 2022 – Dec. 2022 <i>Texas, US</i>
Grader, Math 304 (Linear Algebra) <i>Texas A&M University</i>	Jun. 2022 <i>Texas, US</i>
Grader, Math 423 (Linear Algebra II) <i>Texas A&M University</i>	Jan. 2022 – May. 2022 <i>Texas, US</i>
Grader, Math 467 (Modern Geometry) <i>Texas A&M University</i>	Aug. 2021 – Dec. 2021 <i>Texas, US</i>
Teaching Assistant(Calculus I, Calculus II) <i>Ulsan National Institute of Science and Technology</i>	Mar. 2018 – Dec. 2019 <i>Ulsan, Korea</i>
Calculus I Open Lecture <i>Ulsan National Institute of Science and Technology</i>	Jun.2018 <i>Ulsan, Korea</i>

MENTORING

Direct Reading Program - A first course in Algebraic Topology <i>Texas A&M University</i>	Sep. 2025 - Dec.2025 <i>Texas, US</i>
---	--

SERVICE

Treasurer <i>Association for Women in Mathematics, Texas A&M University</i> • Women in Math Mentoring Lunch • AWM peer mentoring program	Aug.2022 - May.2023 <i>Texas, US</i>
Teaching Assistant for Math Circle <i>Texas A&M Math Circle</i>	Nov. 2023 <i>Texas, US</i>
Grader for High School Math Contest <i>Texas A&M High School Mathematics Contest</i>	Nov. 2023 <i>Texas, US</i>
Grader for High School Math Contest <i>Texas A&M High School Mathematics Contest</i>	Nov. 2022 <i>Texas, US</i>